Content-Based Healthcare Data Classification for Medial Images using Deep Learning

A Project Report submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING

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2026

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DECLARATION

I hereby declare that the project report entitled **"Content-Based Healthcare Data Classification for Medical Images using Deep Learning"** is an original work done in the Department of Computer Science and Engineering, GITAM School of CSE, GITAM (Deemed to be University) submitted in partial fulfillment of the requirements for the award of B. Tech Degree in Computer Science and Engineering. The work has not been submitted to any other college or University for the award of any degree or diploma.

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CERTIFICATE

This is to certify that the project report entitled **"Content-Based Healthcare Data Classification for Medical Images using Deep Learning"** is a bonafide record of work carried out by **Ronda Cholan Reddy (HU22CSEN0100062), Veganti Sahith (HU22CSEN0101590), Gadu Vishnu (HU22CSEN0101639), Anirudh Kotikalapudi (HU22CSEN0100610),** students of this department submitted in partial fulfillment of requirements for the award of B.Tech Degree in Computer Science and Engineering.

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ACKNOWLEDGEMENT

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My project would not have been successful without the help of several people. We would like to thank the personalities who were part of our project in numerous ways, those who gave us outstanding support from the birth of the project.

We are extremely thankful to our honorable Pro-Vice Chancellor, **Dr. D. Shambasiva Rao** for providing necessary infrastructure and resources for the accomplishment of our project.

We are highly indebted to **Prof. S Arun Kumar**, Dean, School of CSE, for his support during the tenure of the project.

We are very much obliged to **Dr. S. Mahaboob Basha**, Head of the Department of Artificial Intelligence and Data Science for providing the opportunity to undertake this project and encouragement in completion of this project.

We hereby wish to express our deep sense of gratitude to **Dr. Naushad Varish,** Assistant Professor, Department of Computer Science and Engineering, School of CSE for the esteemed guidance, moral support and invaluable advice provided by her for the success of the project.

We are also thankful to all the staff members of the Computer Science and Engineering department who have cooperated in making our project a success. We would like to thank all our parents and friends who extended their help, encouragement and moral support either directly or indirectly in our project work.

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TABLE OF CONTENTS

.

ABSTRACT.....6

INTRODUCTION.....7

LITERATURE REVIEW.....8

PROBLEM IDENTIFICATION AND OBJECTIVES.....9

Problem Identification.....9

Objectives.....9

EXISTING SYSTEM..... 10

PROPOSED SYSTEM 12

SYSTEM ARCHITECTURE..... 14

Overview of System Architecture.....14

Input MRI Image.....14

Image Preprocessing.....14

Data Augmentation.....14

Feature Extraction using VGG16.....15

Classification Layer.....15

Output Prediction.....15

User Interface.....15

SYSTEM ARCHITECTURE DIAGRAM..... 16

TOOLS / TECHNOLOGIES USED 17

IMPLEMENTATION AND RESULTS..... 19

Implementation.....19

Results.....20

Interface.....21

CONCLUSION 22

REFERENCES..... 23

ABSTRACT

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Brain tumors are one of the most critical health conditions requiring early detection and accurate classification for effective treatment. This project presents a deep learning-based approach for automatic brain tumor classification using Magnetic Resonance Imaging (MRI) images. A pretrained Convolutional Neural Network, VGG16, is utilized as the base model for feature extraction, enabling the system to learn complex patterns from medical images efficiently. To enhance model performance and reduce overfitting, data augmentation techniques such as rotation, flipping, and zooming are applied to increase dataset diversity.

The proposed system classifies brain MRI images into four categories: glioma, meningioma, pituitary tumor, and normal. The model is trained and evaluated on a labeled dataset and demonstrates high accuracy in distinguishing between different tumor types. Additionally, a user-friendly interface is developed using Gradio, allowing users to upload MRI images and obtain real-time predictions.

The developed system provides a fast, accurate, and efficient solution for brain tumor classification, assisting medical professionals in diagnosis and decision-making. This project highlights the potential of deep learning techniques in improving computer-aided diagnosis systems and contributing to advancements in healthcare.

INTRODUCTION

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Brain tumors are abnormal growths of cells within the brain that can be life-threatening if not detected and treated at an early stage. They can be classified into different types such as glioma, meningioma, and pituitary tumors, each requiring specific medical treatment [1]. Magnetic Resonance Imaging (MRI) is one of the most widely used techniques for detecting brain abnormalities due to its high resolution and ability to capture detailed images of brain tissues [2].

Traditionally, brain tumor diagnosis is performed manually by radiologists through visual inspection of MRI scans. However, this process is time-consuming, requires expert knowledge, and is prone to human error, especially when dealing with large volumes of medical data. Therefore, there is a growing need for automated systems that can assist in accurate and efficient diagnosis [3].

With the advancement of Artificial Intelligence (AI) and deep learning, automated medical image analysis has become more effective. In particular, Convolutional Neural Networks (CNNs) have shown remarkable performance in image classification tasks by automatically extracting important features from images without the need for manual feature engineering [4]. In this project, a pretrained VGG16 model is used as the base architecture to classify brain MRI images into multiple tumor categories [5].

To further enhance the performance of the model, data augmentation techniques such as rotation, flipping, and zooming are applied to increase the diversity of the training dataset and reduce overfitting. This improves the model's ability to generalize to new and unseen data [6].

The proposed system aims to develop an efficient and reliable brain tumor classification model that can assist healthcare professionals in making faster and more accurate decisions. Additionally, a user-friendly interface is developed using Gradio, allowing users to upload MRI images and receive real-time predictions. This integration of deep learning with medical imaging demonstrates the potential of AI in improving diagnostic accuracy and supporting modern healthcare systems [7].

LITERATURE REVIEW

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Brain tumor classification using Magnetic Resonance Imaging (MRI) has become a significant area of research in medical image analysis due to the need for accurate and early diagnosis. Traditional Machine Learning (ML) techniques such as Support Vector Machines (SVM), k-Nearest Neighbors (KNN), and Decision Trees were initially used for tumor detection and classification [1]. These approaches relied heavily on handcrafted feature extraction methods, including texture and intensity-based features. However, such methods often fail to capture complex patterns and variations present in medical images, leading to limited accuracy.

With the emergence of deep learning, Convolutional Neural Networks (CNNs) have demonstrated superior performance in image classification tasks by automatically learning hierarchical feature representations from raw image data [2]. CNN-based approaches eliminate the need for manual feature extraction and provide improved robustness in handling variations in tumor shape, size, and location.

Among various deep learning models, pre-trained architectures such as VGG16 have been widely used through transfer learning to enhance classification performance [3]. Transfer learning allows the model to leverage knowledge gained from large datasets, thereby improving accuracy and reducing training time, especially when medical datasets are limited.

In addition, data augmentation techniques such as rotation, flipping, zooming, and scaling are commonly applied to increase the diversity of training data and prevent overfitting [4]. These techniques help improve the generalization capability of the model and enhance overall performance.

Recent studies have shown that combining deep learning models with effective preprocessing and augmentation strategies significantly improves classification accuracy [5]. Despite these advancements, challenges such as dataset imbalance and variability in tumor characteristics still remain.

PROBLEM IDENTIFICATION AND OBJECTIVES

Problem Identification:

Brain tumors are a serious medical condition that require early and accurate diagnosis for effective treatment. Magnetic Resonance Imaging (MRI) is widely used for detecting brain tumors, but manual analysis of MRI scans by radiologists is time-consuming, subjective, and prone to human error. In many cases, the increasing number of patients and limited availability of medical experts can delay diagnosis, which may lead to severe health risks.

Traditional methods for brain tumor classification rely on manual feature extraction and conventional machine learning techniques, which often fail to capture complex patterns present in medical images. Additionally, variations in tumor size, shape, and location make accurate classification challenging. These limitations highlight the need for an automated and reliable system that can assist in detecting and classifying brain tumors efficiently.

The main problem addressed in this project is the development of an automated brain tumor classification system using deep learning techniques to improve accuracy, reduce human effort, and support medical professionals in decision-making.

Objectives

1. To develop an automated system for classifying brain tumors from MRI images using deep learning models such as VGG16.
2. To improve classification accuracy by applying data augmentation techniques that enhance the diversity of training data.
3. To reduce dependency on manual diagnosis by providing a fast and reliable computer-aided diagnostic system.
4. To assist medical professionals in early detection and classification of brain tumors, enabling timely treatment.
5. To design a system that is efficient, scalable, and capable of handling variations in tumor characteristics such as size, shape, and location.

EXISTING SYSTEM

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In the existing system, brain tumor detection and classification are primarily carried out through manual examination of Magnetic Resonance Imaging (MRI) scans by radiologists and medical experts. The radiologist visually inspects the MRI images to identify abnormal regions and determine the type and severity of the tumor based on experience and clinical knowledge. Although this approach is widely adopted in hospitals and diagnostic centers, it is highly dependent on the expertise of the individual practitioner.

As a result, the diagnostic outcome may vary from one radiologist to another, leading to inconsistencies and subjectivity in interpretation. In addition, the manual inspection process is time-consuming, especially when dealing with a large number of patient cases, which can delay diagnosis and treatment [1]. The growing demand for medical imaging analysis, combined with a limited number of skilled professionals, further increases the burden on healthcare systems.

To address some of these limitations, traditional machine learning methods have been introduced for automated brain tumor classification. These systems typically involve multiple stages, including image preprocessing, feature extraction, feature selection, and classification. In such approaches, features like texture, intensity, shape, and statistical measures are manually extracted from MRI images using techniques such as Gray-Level Co-occurrence Matrix (GLCM) and histogram-based methods.

These features are then fed into classifiers such as Support Vector Machines (SVM), k-Nearest Neighbors (KNN), and Decision Trees to categorize tumors into different classes. While these methods provide some level of automation, they heavily rely on handcrafted features, which require domain expertise and significant effort to design. Moreover, these manually extracted features may not fully represent the complex and nonlinear patterns present in medical images, leading to limited performance and reduced accuracy [2].

Another major challenge in existing systems is their inability to effectively handle the high variability and complexity of brain tumors. Tumors can differ significantly in terms of size, shape, texture, and location within the brain. In some cases, tumors may appear very similar to normal tissues, making them difficult to detect accurately. Traditional methods often struggle to generalize across such diverse cases, especially when trained on limited datasets.

This lack of generalization can result in misclassification and unreliable predictions, which are critical issues in medical diagnosis [3]. Furthermore, many datasets used in earlier systems are relatively small and may suffer from class imbalance, where certain tumor types are underrepresented. This imbalance can bias the model toward dominant classes, reducing its effectiveness in real-world scenarios.

In addition to accuracy-related challenges, existing systems also face issues related to efficiency and scalability. Manual and semi-automated approaches involve multiple processing steps that require human intervention, making them unsuitable for large-scale deployment. As the volume of medical imaging data continues to grow, these systems struggle to process data quickly and efficiently.

This limitation can lead to delays in diagnosis, which is particularly critical in cases where early detection is essential for successful treatment. Moreover, the computational efficiency of traditional machine learning models is often limited when handling high-dimensional image data, further affecting performance [4].

Another important limitation is the lack of full automation in the existing systems. Most approaches require human involvement at various stages, including preprocessing of images, selection of features, and validation of results. This dependency reduces the overall reliability and consistency of the system and increases the possibility of human error. Additionally, such systems are not easily scalable to handle real-time clinical applications or large hospital databases. The absence of robust automation also limits their integration into modern healthcare systems, where rapid and accurate decision-making is essential.

Overall, while existing methods provide a foundation for brain tumor detection and classification, they suffer from several critical limitations, including dependency on manual analysis, reliance on handcrafted features, limited ability to capture complex patterns, poor generalization across diverse tumor types, and lack of scalability. These challenges highlight the need for more advanced, fully automated approaches that can improve accuracy, reduce processing time, and provide consistent and reliable results for clinical use [5].

PROPOSED SYSTEM

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The proposed system presents a comprehensive and fully automated framework for brain tumor classification using advanced deep learning techniques. The main objective of this system is to overcome the limitations of traditional diagnostic approaches, such as manual analysis and conventional machine learning methods, by leveraging the powerful feature learning capabilities of Convolutional Neural Networks (CNNs). In this system, brain MRI images are directly processed by a deep learning model, which automatically learns relevant patterns and predicts the type of tumor without requiring manual intervention. This significantly reduces dependency on human expertise, minimizes diagnostic errors, and improves the overall efficiency and reliability of the classification process.

The core of the proposed system is based on a pretrained **VGG16**, which is a deep CNN architecture consisting of multiple convolutional and fully connected layers. VGG16 is widely known for its simplicity and effectiveness in image classification tasks due to its uniform architecture using small 3×3 convolution filters. Instead of training the model from scratch, transfer learning is employed, where the pretrained weights of VGG16, originally trained on large-scale datasets such as ImageNet, are reused. This approach enables the model to leverage previously learned visual features, allowing it to perform well even with limited medical image data. The early convolutional layers capture basic features such as edges and textures, while deeper layers extract more complex and abstract representations, which are essential for distinguishing between different tumor types.

In the proposed system, the pretrained VGG16 model is modified to suit the specific task of brain tumor classification. The original fully connected layers of VGG16 are replaced with custom classification layers tailored to the dataset. Typically, a Global Average Pooling or Flatten layer is added after the convolutional base, followed by one or more dense layers with activation functions such as ReLU. Dropout layers are also incorporated to prevent overfitting by randomly deactivating neurons during training. The final output layer uses a softmax activation function to generate probability scores for each tumor class, ensuring that the model provides a confident and interpretable prediction.

To address the challenge of limited dataset size and improve model generalization, data augmentation techniques are extensively applied during the training phase. Medical imaging datasets are often small and may not cover all variations of tumor appearance, which can lead to overfitting. To mitigate this issue, transformations such as rotation, horizontal and vertical flipping, zooming, shifting, and brightness adjustments are applied to the training images.

These transformations create multiple variations of the same image while preserving its medical significance. As a result, the model is exposed to a wider range of patterns and becomes more robust when handling unseen data.

Another important aspect of the proposed system is preprocessing, which ensures that all input images are standardized before being fed into the model. MRI images are resized to a fixed dimension of 224×224 pixels to match the input requirements of the VGG16 architecture. Additionally, pixel values are normalized to a standard range, which improves numerical stability and accelerates the training process. Preprocessing helps maintain consistency across the dataset and ensures optimal performance of the deep learning model.

The system is specifically designed to classify brain MRI images into four distinct categories: glioma, meningioma, pituitary tumor, and normal. These categories represent common types of brain tumors and normal brain conditions. By accurately distinguishing between these classes, the system can assist in early detection and diagnosis, which is critical for effective treatment planning.

To make the system practical and accessible, a user-friendly interface is developed using the Gradio framework. This interface allows users to upload MRI images through a simple web-based platform and receive predictions in real time. The output is displayed in an intuitive format, showing the predicted tumor type along with probability scores for each class. This feature enhances usability and makes the system suitable for both technical users and healthcare professionals who may not have a background in machine learning.

Furthermore, the proposed system is designed to be scalable and efficient, making it suitable for real-world deployment. The use of transfer learning reduces computational requirements and training time, while the automated pipeline ensures fast processing of input images. The system can be easily extended to include additional tumor types or integrated with hospital information systems for large-scale clinical applications.

Overall, the proposed system provides a robust, accurate, and efficient solution for brain tumor classification. By combining deep learning, transfer learning, data augmentation, and an interactive user interface, the system addresses the limitations of existing approaches and offers significant improvements in diagnostic accuracy and speed. This demonstrates the potential of artificial intelligence in transforming medical image analysis and supporting modern healthcare systems in delivering better patient outcomes.

SYSTEM ARCHITECTURE

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Overview of System Architecture

The system architecture describes the overall workflow of the brain tumor classification system, starting from input MRI images to the final prediction output. The system follows a structured pipeline where images are first preprocessed and enhanced using data augmentation techniques. These processed images are then passed through a pretrained VGG16 model for feature extraction. The extracted features are further processed by classification layers to determine the tumor type. Finally, the system provides the predicted result through a user-friendly interface, enabling real-time interaction.

Input MRI Image

The process begins with the input of brain MRI images collected from the dataset. These images belong to different categories such as glioma, meningioma, pituitary tumor, and normal. The quality and diversity of these images play an important role in model performance.

Image Preprocessing

Before feeding the images into the model, preprocessing is performed to ensure uniformity. The images are resized to 224×224 pixels to match the input requirements of the model. Pixel values are normalized to bring them into a standard range, which helps in faster convergence during training.

Data Augmentation

Data augmentation techniques are applied to increase the diversity of the dataset. Operations such as rotation, flipping, and zooming are performed on images to create multiple variations. This helps the model learn better and reduces the chances of overfitting.

Feature Extraction using VGG16

The preprocessed images are passed through the pretrained VGG16 model. The convolutional layers of VGG16 extract important features such as edges, shapes, and textures from the images. This step eliminates the need for manual feature extraction and improves efficiency.

Classification Layer

The extracted features are then passed to fully connected layers, which perform the classification task. These layers analyze the features and assign the image to one of the tumor categories based on learned patterns.

Output Prediction

The final output is generated using a softmax function, which produces probability values for each class. The class with the highest probability is selected as the final prediction.

User Interface (Gradio)

The system includes a user-friendly interface built using Gradio. Users can upload MRI images, and the system processes them and displays the predicted tumor type instantly. This makes the system easy to use and suitable for real-time applications.

SYSTEM ARCHITECTURE DIAGRAM

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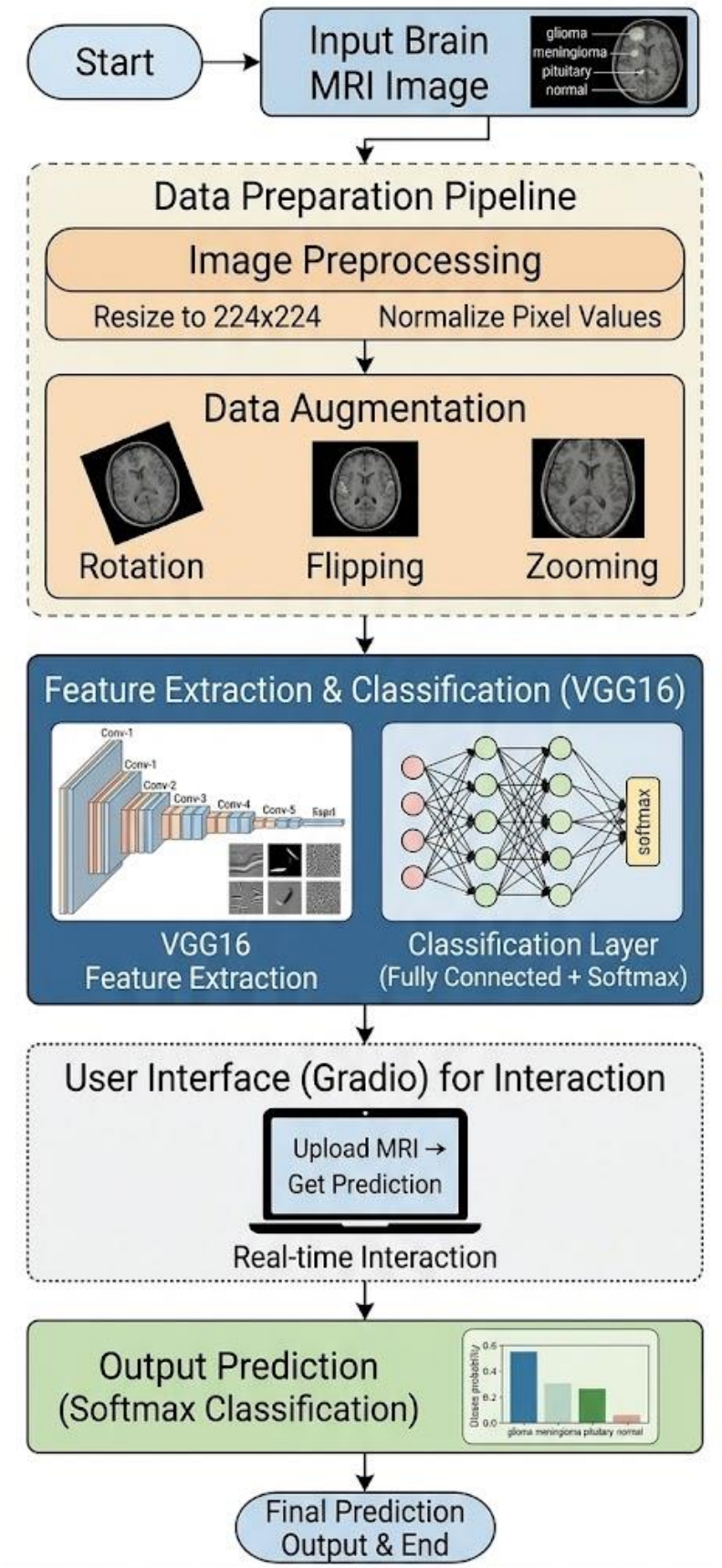

TOOLS / TECHNOLOGIES USED

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Programming Language

The entire system is developed using Python, which is widely used in machine learning and deep learning applications due to its simplicity and rich library support.

Deep Learning Framework

The project uses TensorFlow and Keras for building and training the deep learning model. TensorFlow provides a powerful backend for computations, while Keras offers a user-friendly interface for designing neural networks.

Pretrained Model

A pretrained VGG16 model is used as the base architecture. It helps in extracting important features from MRI images using transfer learning, improving accuracy and reducing training time.

Image Processing

OpenCV is used for image processing tasks such as reading images, resizing, and basic transformations. It plays an important role in preparing the input data for the model.

Data Handling

NumPy is used for handling numerical data and performing array operations efficiently. It helps in converting images into arrays and managing datasets.

Data Augmentation

The Image Data Generator module from Keras is used for applying data augmentation techniques such as rotation, flipping, and zooming. This improves model performance and reduces overfitting.

Visualization

Matplotlib is used for visualizing results such as images, predictions, and model outputs during testing and evaluation.

User Interface

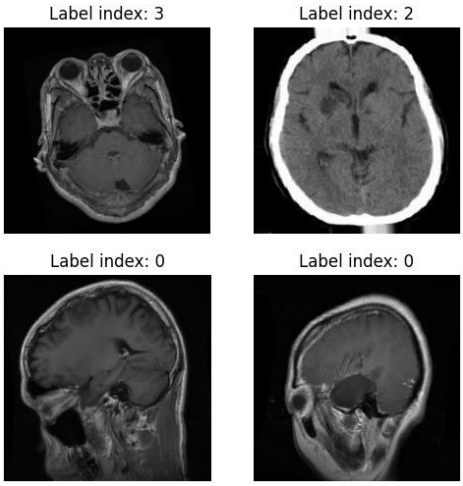
Gradio is used to build a simple and interactive web interface. It allows users to upload MRI images and get real-time predictions, making the system easy to use.

IMPLEMENTATION AND RESULTS

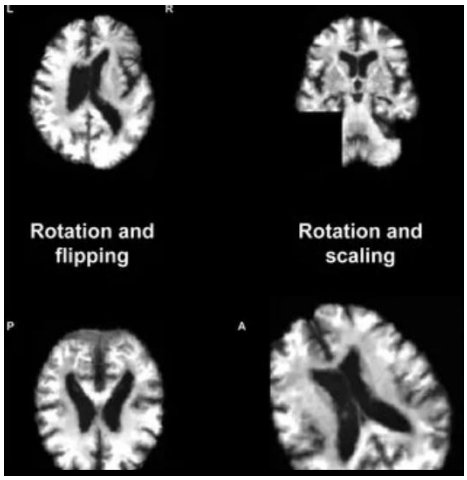
IMPLEMENTATION:

Data Preprocessing& Data Augmentation:

The MRI images are first preprocessed to ensure consistency in input data. All images are resized to a standard size of 224×224 pixels to match the input requirements of the model. The pixel values are normalized to a range of 0 to 1, which helps in faster and more stable training. This step ensures that all images have uniform dimensions and intensity values before being passed to the model.

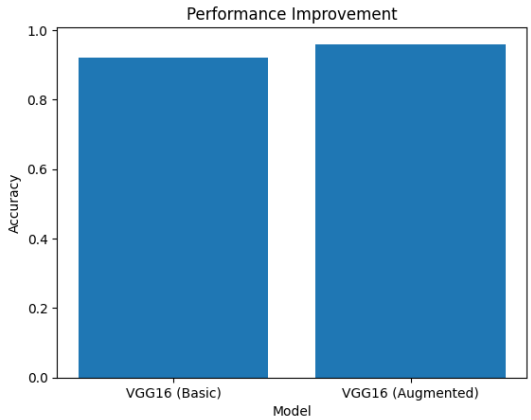


To improve the performance of the model and reduce overfitting, data augmentation techniques are applied to the training dataset. Transformations such as rotation, horizontal flipping, and zooming are used to create multiple variations of the same image. This increases the diversity of the dataset and helps the model learn more robust features, improving its ability to generalize on unseen data.



Model training:

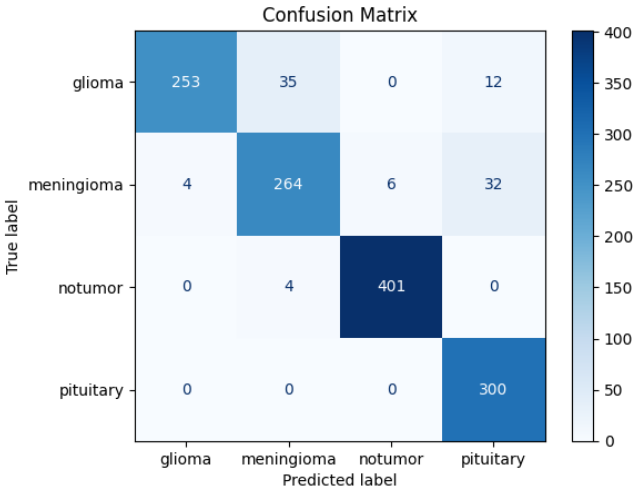
A pretrained VGG16 model is used for feature extraction. The convolutional layers of VGG16 extract important features from MRI images, and additional fully connected layers are added for classification. The model is trained using the augmented dataset, and performance is evaluated using metrics such as accuracy and loss. The trained model is then used for predicting tumor classes.



RESULTS:

Confusion Matrix Analysis:

The confusion matrix shows that the VGG16 model performs very well for no tumor and pituitary classes with high correct predictions. However, there is some confusion between glioma and meningioma, where a few cases are misclassified due to similar visual features. Overall, the model demonstrates strong classification performance with minor errors in specific classes.



Accuracy Comparison:

The accuracy graph shows that both training and validation accuracy increase steadily, reaching around 94–95%, indicating good learning. The small gap between them shows that the model

generalizes well without significant overfitting. The model performs best for no tumor and pituitary, while meningioma has slightly lower performance.

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41/41 ━━━━━━━━━━━━━━━━━━━━ 7s 162ms/step
--- ADVANCED EVALUATION METRICS ---
Tumor Type      Precision    Recall      F1-Score    Support
glioma          0.98444      0.84333     0.90844     300
meningioma      0.87129      0.86275     0.86700     306
notumor         0.98526      0.99012     0.98768     405
pituitary       0.87209      1.00000     0.93168     300
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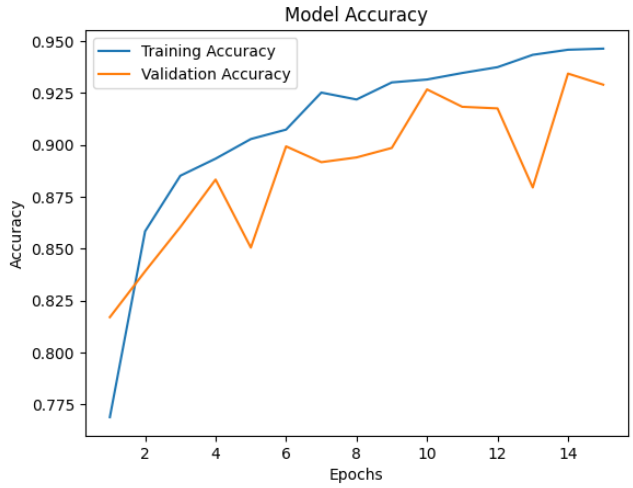
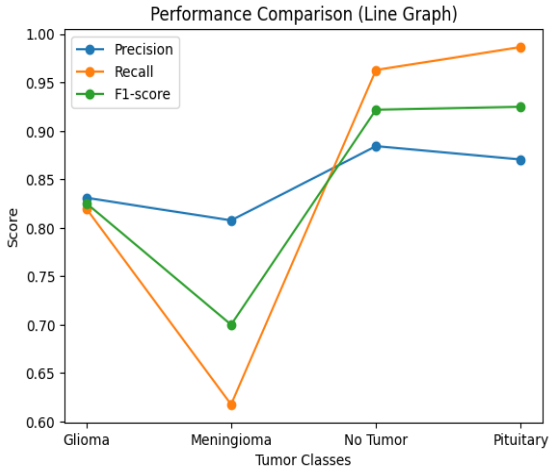
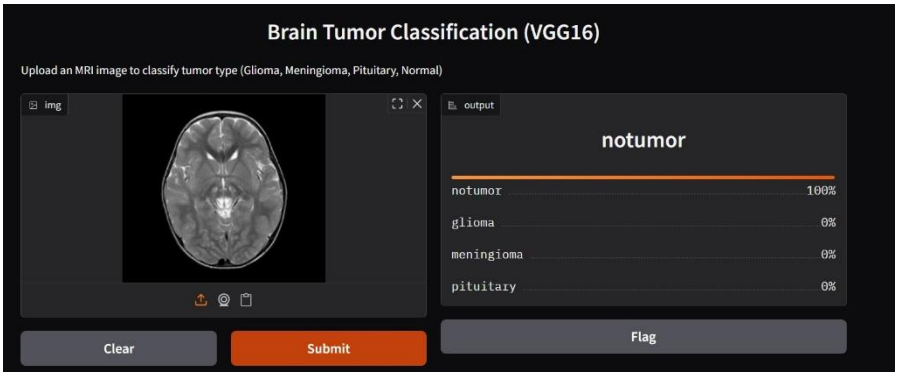




Final Outcome:

The proposed system achieved high accuracy of approximately 94–95% using the VGG16 model for brain tumor classification. It successfully classifies MRI images into four categories: glioma, meningioma, pituitary, and no tumor. The model demonstrates good generalization on unseen data, indicating its robustness and reliability. Additionally, the system is integrated with a Gradio interface, allowing users to upload MRI images and obtain real-time predictions along with probability scores for each class. This makes the system user-friendly and easy to use. Overall, the developed system is efficient, accurate, and suitable for supporting clinical decision-making.



CONCLUSION

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This project presents an effective approach for brain tumor classification using deep learning techniques applied to MRI images. A pretrained **VGG16** model is used to automatically extract important features, improving accuracy and reducing training time through transfer learning.

Data augmentation techniques such as rotation, flipping, and zooming enhance the model's ability to generalize and prevent overfitting. The system successfully classifies MRI images into four categories: glioma, meningioma, pituitary tumor, and normal, achieving reliable performance.

Additionally, the integration of a Gradio-based interface allows users to upload images and obtain real-time predictions, making the system user-friendly and practical. Overall, the proposed system provides an accurate, efficient, and automated solution that can assist medical professionals in faster and more reliable diagnosis.

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